Feature Selection and Exclusion Rationale

(Documented in Notebook Commenting for source code file Machine Learning Model.ipynb)

Feature Selection

The feature-selection process uses a **canonical feature ownership** approach. Each feature concept has exactly one canonical source, and equivalent attributes appearing in multiple dimensions are taken from the single owning dimension only.

The code defines the canonical ownership as:

- Demographics and student status — fact table
- EFTSL and cumulative credit points — fact table
- Commencing/continuing status — fact table
- Teaching-period and calendar attributes — teaching-period dimension
- Course group, course level and primary field of education — course dimension
- Organisation attributes — excluded because no populated attributes are available
- Offering-specific attributes — excluded because no populated attributes are available

The 12 source tables are profiled column by column to determine which candidate attributes are populated and usable. The notebook records that the resulting **21 assembled features represent complete coverage of the usable STAR schema** in the current synthetic generation.

Final 21 Features

Demographics and student status

1. age_at_census — student age at the census-date snapshot.
2. socioeconomic_status — SEIFA quartile recorded at census date.
3. regional_remote_status — ASGS remoteness recorded at census date.
4. student_gender — student gender.
5. student_is_international_student — canonical international status.
6. student_is_first_nations_student — First Nations status.

Enrolment and study load

7. attendance_mode — attendance mode group at the enrolment snapshot.
8. eftsl — equivalent full-time student load for the enrolment.
9. commencing_continuing — canonical commencing/continuing concept for the calendar year.

10. commencing_continuing_period — teaching-period commencing status. It differs from the calendar-year value on 36% of rows and is therefore treated as distinct information rather than a duplicate encoding.
11. course_admission_load_category — full-time/part-time course-admission load. It is distinct from attendance_mode and fully populated.
12. enrolment_year — enrolment year derived from the census date.

Cumulative academic outcomes

13. cumulative_credit_points_enrolled — cumulative load up to census date.
14. cumulative_credit_points_passed — cumulative passes up to census date.
15. cumulative_credit_points_failed — cumulative failures up to census date.
16. cumulative_credit_points_withdrawn — cumulative withdrawals up to census date.

Course characteristics

17. course_group — course group owned by the course dimension.
18. broad_primary_field_of_education — canonical broad field of education.
19. narrow_primary_field_of_education — canonical narrow field of education.
20. detailed_primary_field_of_education — canonical detailed field of education.

Teaching-period characteristics

21. teaching_period — teaching period owned by the teaching-period dimension.

The executed validation records that all 21 approved features are **100% populated with zero missingness**.

Feature Exclusion Rationale

Attrition leakage variables

is_twelve_month_course_attrition is excluded because it is a prohibited leakage column.

The executed data also confirms that every row where is_twelve_month_course_attrition = false has is_twelve_month_student_attrition = false. This represents 747,744 rows, meaning course attrition alone would identify 76.8% of the population as guaranteed negatives.

The following are also excluded:

- is_tcsi_student_attrition
- is_tcsi_course_attrition
- is_withdrawn_student_attrition
- is_withdrawn_course_attrition

These are documented as post-outcome attrition indicators and are also all-NULL in the current generation.

Outcome-derived identifiers

student_attrition_deidentified_hash and course_attrition_deidentified_hash are excluded because they are outcome-derived hashes. The feature manifest states that identifiers are never predictors.

Raw keys and hashes

All *_key and *_key_hash columns are excluded because raw keys and hashes are used for joins and traceability only.

age_band

age_band is excluded because it is a banded duplicate of age_at_census. The numeric representation is retained as canonical.

Duplicate international-status fields

The following are excluded:

- is_international_student
- international_domestic_enrolment
- international_domestic_student

They are documented as exact duplicates of the canonical student_is_international_student.

Duplicate commencing/continuing fields

The following are excluded:

- is_commencing
- is_commencing_12m
- is_commencing_period
- is_commencing_half_year
- commencing_continuing_12m
- commencing_continuing_half_year

They are documented as boolean or duplicate encodings of the two retained commencing features.

commencing_continuing_12m equals commencing_continuing on 100% of rows.

Zero-variance fields

is_enrolment and is_study_load are excluded because they are populated but single-valued and constant true, so they carry zero variance.

Census-date fields

census_date and course_enrolment_census_date are excluded because they are snapshot reference dates rather than approved predictive features. enrolment_year already carries the calendar signal.

course_level

course_level is excluded because it contains the same values as course_group in the current generation.

calendar_year

calendar_year is excluded because it duplicates the fact-owned enrolment_year.

Study Area A and Study Area B field-of-education fields

Field-of-education attributes from Study Area A and Study Area B are excluded.

Both dimensions join one-to-one and contain populated attributes, but they duplicate the field-of-education concept owned by the course dimension. The course dimension remains the single canonical field-of-education owner.

Unit field-of-education fields

Unit field-of-education attributes are excluded because they duplicate the field-of-education concept and because the unit key was not unique in the tested generation.

The notebook records that joining the unit dimension would inflate the fact table from **973,770 to 7,508,854 rows**.

Course-offering and unit-offering attributes

These attributes are excluded because no populated non-key attributes are available in the current generation.

Organisation attributes

Organisation attributes are excluded because no populated non-key attributes are available.

Module, module-offering and thesis attributes

These are excluded because curriculum_item_key_hash maps to unit, module and thesis in the relationship model, while the fact keys resolve only to unit. These dimensions are therefore profiled but not joined.

Remaining fact-table columns

All remaining fact-table columns are excluded where they are completely NULL in the current synthetic generation.

The source audit records that 38 further fact-table columns returned zero non-null rows.

Final Feature Validation

The code verifies that the final feature set contains no:

- target field;
- prohibited leakage field;
- identifier;
- raw key;

- key hash; or
- other hash.

The feature-vector validation performs the same check again before model training and confirms that all approved features are present.

Any approved feature that becomes completely NULL after assembly can be removed as a validation correction. In the executed dataset, none of the 21 final features required removal because all were fully populated.

Generated using OpenAI with prompt:

Hello, I have attached the source code file for the Machine Learning Model notebook (ipynb). I need you to read through this file and extract the Feature Selection and Exclusion Rationale that is already documented in the source code.

Investigate the notebook comments, markdown sections, feature manifest, validation checks and documented design decisions that relate directly to why features were included or excluded.

- The final 21 features used by the model and why they were kept.
- Where the main features come from in the dataset.
- Which features were removed and why they were removed.
- Include reasons such as duplicate information, missing data, identifiers, data leakage or features that could not be safely used.
- Include any evidence in the notebook that supports these decisions, such as percentages, row counts or duplicate checks.
- Include the final checks used to make sure only the approved features are used by the model.
- Only use information and reasoning that is directly documented in the attached notebook file. Do not add external machine learning reasoning or create new justification that is not present in the source code.
- Summarise and organise the extracted information so it can be copied into development documentation under the heading "Feature Selection and Exclusion Rationale".

If a feature does not have a documented reason for inclusion or exclusion, state this rather than creating a reason.